# Face variation: Reputation-based Reward System & Manual Validation Framework

# Post-Analysis of Phase 4 — Cycle 5 Execution Phase (Oct 22, 2026)

Phase 4, Cycle c5(Post-Analysis) – Oct 22, 2026

### Summary: Why Scores Dropped — Uniqueness, Not Identity

A large share of the score reductions seen in Phase 4 — Cycle 5 came from the similarity (duplicate) check, not from failing the variation task itself. Many miners are distinct wallets, with different coldkeys and hotkeys, yet they are submitting near-identical images: same overall look. That pattern almost always points to the same, or a very similar, generation model combined with the same prompt recipe.

The goal of this network is to collect unique training data. We do not want the same photo submitted 100 times, and a dataset full of repeated images has very little value. The similarity gate is intentionally strict for that reason.

**What miners should do:** change the model and/or the prompt enough that your outputs are visibly different — and different in embedding space — from other miners working on the same challenge, even when everyone starts from the same seed face.

### What the Similarity Check Is Measuring

A submission can pass validation with a high validation score and still receive a similarity penalty. The two checks measure different things:

| Check | Typical result | What it means |
|---|---|---|
| Variation / identity task | Often passes | You performed the edit that was requested. |

| Check | Typical result | What it means |
|---|---|---|
| Similarity / duplicate check | Fails | Your image is too close to another miner's image on the same challenge and variation slot. |

In other words, the Miner may be unique, but the pixels and embeddings are not.

## What We Found in Two Local Challenges

We re-ran the production similarity detector (image statistics) on two challenges from Phase 4 — Cycle 5. Results across those two challenges:

| Metric | Value |
|---|---|
| Cross-coldkey flagged pairs | 688 |
| Distinct coldkeys involved in those pairs | 39 |

This is the key point: dozens of separate miners are colliding with each other, not one repeated hotkey. As one illustration, a single miner in these packages had 6 of its 8 images flagged, involving 51 flagged pairs with 25 different partner hotkeys belonging to 10 different coldkeys. Every one of those partners is another miner. Same seed, same slot and same look results in a similarity penalty.

Note: this is evidence from two challenges. The mechanism and the visual failure mode generalize.

## Side-by-Side Examples (All Different Miners)

Each pair below shows two submissions from two different coldkeys on the same challenge and variation slot. Match strength shows how far the pair falls under the variation cut: strength % = (1 − combined distance ÷ threshold) × 100, bounded between 0 and 100.

| # | Variation | Distance | Threshold | Strength | Challenge |
|---|---|---|---|---|---|
| 1 | background_in | 0.019 | 0.20 | ~90% | …5Dvgtk1b |

| # | Variation | Distance | Threshold | Strength | Challenge |
|---|---|---|---|---|---|
| 2 | background_out | 0.001 | 0.20 | ~99.5% | …5Dvgtk1b |
| 3 | lighting_edit + expression_edit | 0.028 | 0.10 | ~72% | …5Dvgtk1b |
| 4 | lighting_edit + pose_edit | 0.018 | 0.08 | ~77% | …5GWzXSra |
| 5 | pose_edit + expression_edit | 0.027 | 0.06 | ~56% | …5GWzXSra |

## Example 1 — background_in

Distance 0.019 against a background threshold of 0.20 (~90% strength).



*Miner A*



*Miner B*

## Example 2 — background_out

Distance 0.001 against a threshold of 0.20 (~99.5% strength).



*Miner A*



*Miner B*

## Example 3 — lighting_edit + expression_edit

Distance 0.028 against a threshold of 0.10 (~72% strength).



*Miner A*



*Miner B*

### Example 4 — lighting_edit + pose_edit

Distance 0.018 against a threshold of 0.08 (~77% strength). Different challenge from Examples 1–3.



*Miner A*



*Miner B*

### Example 5 — pose_edit + expression_edit

Distance 0.027 against a threshold of 0.06 (~56% strength).



*Miner A*



*Miner B*

### Example 6 — Many miners, one look (background_in)

A single hub image (top) matched 18 other images on background_in within one challenge, across several different wallets. Six of those partner images are shown below. We do not need this look submitted many times.



*Hub image*



*Partner 1* *Partner 2* *Partner 3* *Partner 4* *Partner 5* *Partner 6*

### What Miners Should Do Going Forward

Uniqueness is now part of the job. If your pipeline is “download the seed, run a fixed checkpoint, apply a fixed prompt template,” you will collide with every other miner running the same stack. Starting now, miners should focus on making each image genuinely different:

- **Use different models.** Rotate or upgrade checkpoints, and vary LoRAs and samplers, so your output is not the default look everyone else ships.
- **Improve and vary your prompts.** Change style, framing — not a one-word tweak that still lands in the same embedding neighborhood.
- **Use different accessory variations.** there can be a lot of color and a lot of different variations and hats, religious head coverings etc
- **Use different backgrounds and settings.** Vary surfaces, scenery, time of day and color palette rather than reusing one scene.
- **Vary clothing and hair presentation where the variation allows it.** Do not keep every other detail fixed.
- **Self-check before submitting.** If your outputs look interchangeable with a neighbor's gallery, expect a similarity penalty.

Remember the product goal: diverse, unique samples for the dataset — not 100 copies of one generation. Miners who invest in distinct models and varied prompts will see fewer similarity penalties and will be contributing the data the network actually needs.

# End-Phase 4 — Cycle 5 Updates

Phase 4, Cycle c5(End) – Oct 19th

## Status of Manual Grading

We are officially done with manual review for Phase 4 — Cycle 5, and we are pleased with the results.

For this cycle, our goal was to **collect Real adversarial data for Passive Liveness**. With the data provided through the addresses shared for this cycle, we were able to release **Version 3 of Passive Liveness**

A major highlight from this cycle is the volume of data received. Active hotkeys grew from 159 to 204, and total submissions rose from 1,009,700 to 1,590,992, an increase of roughly 58%. Total challenges also grew, from 3,287 to 4,141. We also got more then **12,000** real screen replay image.

Scoring follows the same flow and criteria used to calculate the predicted scores shared in the last update. Overall, the final results were what was predicted. Where a score dropped from the predicted number, the similarity penalty was the reason. As with previous cycles, miners running more than one hotkey tended to see a larger similarity penalty than single-hotkey miners, which again shows that running multiple hotkeys is not in a miner's best interest.

We appreciate everyone who contributed data during Cycle 5. The data collected this cycle has directly contributed to **the improvement of Passive Liveness]**, and we look forward to building on it in the next cycle.

| Metric | Phase 4 – Cycle 4 | Phase 4 – Cycle 5 |
|---|---|---|
| Hot Keys (Active) | 159 | 204 |
| Total Score | 2,998.13 | 3,500.46 |

| | | |
|---|---|---|
| Total Challenges | 3,287 | 4,141 |
| Total Submissions | 1,009,700 | 1,590,992 |
| Positive Score Rate | 73.65% (780,782 images) | 69.59% (1,978,750 images) |

## Time-Based Rewards and Reputation

We continue to reward miners who have consistently contributed high-quality, non-exploitative data over multiple phases. The bonus structure for Phase 4 — Cycle 5 is:

| Participation History | Age Factor | Bonus |
|---|---|---|
| Phase 3 — Cycle 1 (and active now) | 1.50 | +50% |
| Phase 4 — Cycle 1 (and active now) | 1.40 | +40% |
| Phase 4 — Cycle 2 (and active now) | 1.30 | +30% |
| Phase 4 — Cycle 3 (and active now) | 1.20 | +20% |
| Phase 4 — Cycle 4 (and active now) | 1.10 | +10% |
| Phase 4 — Cycle 5 (new miner) | 1.00 | +0% |

## Final Score Calculation

We're using the same formula as last cycle:

**Total Score = ((Face Variation Score * 0.5) + (Screen replay score * 0.5))× Age Factor**

Where:

- Face Variation Score is computed from the Δrep mapping table (unchanged, see below).
- The real Screen replay's sent by the miners
- Age Factor is based on the participation history described above.

### New Tier (Extended Top-End Normalization)

Note any tier changes this cycle. For reference, the table is included again below:

| Tier | Range of Score | Multiplier |
|---|---|---|
| 🏅 Platinum | 150–249.99 | 1.25 |
| 💎 Diamond | 50–149.99 | 1.15 |
| ⭐ Gold | 10–49.99 | 1.10 |
| 🥈 Silver | 2.00–9.999 | 1.05 |
| 🥉 Bronze | 1.00–1.999 | 1.02 |
| ☐ Neutral | 0.70–1.00 | 1.00 |
| ‼ Watch | 0.10–0.699 | 0.90 |

## Decay

Decay will increase from 0.02-> 0.05

## Leaderboard Growth

Leaderboard growth summary — compare Cycle 5 submissions and positive score rate against Cycle 4.

| Metric | Phase 4 – Cycle 4 | Phase 4 – Cycle 5 |
|---|---|---|
| Total Submissions | 1,009,700 | 1,590,992 |
| Positive Score Rate | 73.65% | 69.59% |

Now that final scoring (including the similarity penalty) is complete, the reputation leaderboard for Phase 4 — Cycle 5 holds. The tier breakdown for this cycle:

| Tier | Cycle 4 | Cycle 5 |
|---|---|---|
| Platinum | 1 | 1 |
| Diamond | 29 | 24 |
| Gold | 17 | 26 |

| | | |
|---|---|---|
| Silver | 26 | 23 |
| Bronze | 37 | 18 |

## Miner's Reputation Leaderboard (Top 10)

Reputation is determined by validated Face performance. Miners not listed here fall into the Neutral tail or the Watchlist, or did not score high enough to surface.

| Rank | Hotkey | Score | Tier |
|---|---|---|---|
| 1 | 5Hmz | 180.12 | Platinum |
| 2 | 5C5h | 129.08 | Diamond |
| 3 | 5GCX | 126.83 | Diamond |
| 4 | 5FCD | 109.10 | Diamond |
| 5 | 5EBx | 104.00 | Diamond |
| 6 | 5CCe | 102.66 | Diamond |
| 7 | 5DJk | 86.12 | Diamond |
| 8 | 5DhK | 72.39 | Diamond |
| 9 | 5DWm | 71.43 | Diamond |
| 10 | 5CorH | 68.37 | Diamond |

## New Δrep Mapping Table

| Validation Score | Meaning | Δrep |
|---|---|---|
| +6 | Breaks system | +0.15 |
| +5 | Perfect | +0.1 |
| +4 | Near Perfect | +0.04 |
| +3 | Good | +0.03 |
| +2 | Okay | +0.02 |
| +1 | Decent | +0.01 |
| 0 | Unclear | 0.00 |
| –1 | Partial spam | -0.05 |

| | | |
|---|---|---|
| –2 | Misleading | -0.08 |
| –3 | Duplicate | –0.10 |
| –4 | Not enough info | -0.25 |
| –5 | Cheat | -0.5 |

## Top 2 Miners' Score Breakdown

The tables below show how raw validation outcomes translated into reputation for the top three miners this cycle.

***5Hmz: Hotkey (Rank 1)***

***Face***

| Score | -5 | -4 | -3 | -2 | -1 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Count | 0 | 0 | 7 | 11 | 146 | 148 | 703 | 715 | 1134 | 556 | 564 | 0 | 3984 |
| Δrep | -0.5 | -0.25 | -0.1 | -0.08 | -0.05 | 0 | 0.01 | 0.02 | 0.03 | 0.04 | 0.1 | 0.15 | |
| Total | 0 | 0 | -0.7 | -0.88 | -7.3 | 0 | 7.03 | 14.3 | 34.02 | 22.24 | 56.4 | 0 | 125.11 |

***Screen replay***

| Score | -5 | -4 | -3 | -2 | -1 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Count** | 0 | 0 | 0 | 0 | 0 | 0 | 72 | 0 | 831 | 821 | 216 | 233 | **2173** |
| Δrep | -0.5 | -0.25 | -0.1 | -0.08 | -0.05 | 0 | 0.01 | 0.02 | 0.03 | 0.04 | 0.1 | 0.15 | |
| **Total** | 0 | 0 | 0 | 0 | 0 | 0 | 0.72 | 0 | 24.93 | 32.84 | 21.60 | 34.95 | **115.04** |

Face score = 125.11

Screen replay score = 115.04

Last cycle score = 0

Age = 6 cycles

Face score + Screen score = (125.11 * 0.5) + (115.04 * 0.5) = 120.08

Age score = 120.08 × 1.6 = 180.12

**Rep score = 180.12**

***5C5h: Hotkey (Rank 2)***

***Face***

| Score | -5 | -4 | -3 | -2 | -1 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Count** | 0 | 0 | 10 | 46 | 186 | 379 | 773 | 889 | 933 | 507 | 207 | 0 | 3930 |
| Δrep | -0.5 | -0.25 | -0.1 | -0.08 | -0.05 | 0 | 0.01 | 0.02 | 0.03 | 0.04 | 0.1 | 0.15 | |
| **Total** | 0 | 0 | -1.0 | -3.68 | -9.3 | 0 | 7.73 | 17.78 | 27.99 | 20.28 | 20.7 | 0 | 80.5 |

***Screen***

| Score | -5 | -4 | -3 | -2 | -1 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Count** | **0** | **0** | **0** | **0** | **0** | **0** | **13** | **3** | **902** | **567** | **338** | **229** | **2052** |
| Δrep | -0.5 | -0.25 | -0.1 | -0.08 | -0.05 | 0 | 0.01 | 0.02 | 0.03 | 0.04 | 0.1 | 0.15 | |
| **Total** | **0** | **0** | **0** | **0** | **0** | **0** | **0.13** | **0.06** | **27.06** | **22.68** | **33.80** | **34.35** | **118.08** |

Face score = 80.5

Screen replay score = 118.04

Last cycle score = 0

Age = 4 cycles

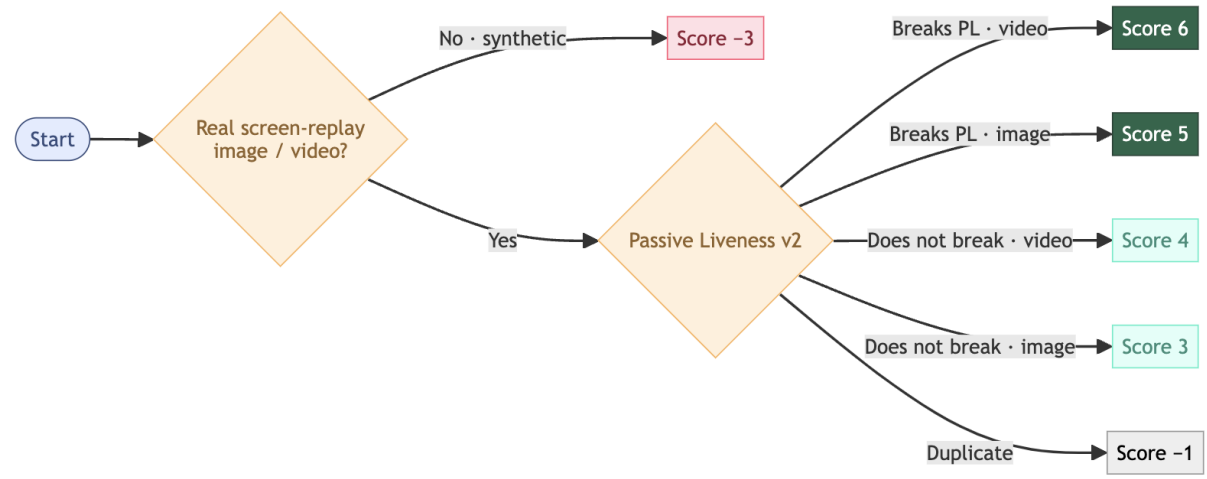
Face score + Screen score = (80.5 * 0.5) + (118.04 * 0.5) = 99.29

Age score = 99.29 × 1.3 = 129.08

**Rep score = 129.08**

# How manual review scoring works

## Flow




## Score sheet

| score | Reason |
|---|---|
| 6 | Video that broke our PL |
| 5 | Image that broke our PL |
| 4 | Video that did NOT break our PL |
| 3 | Image that did not break our PL |

## Issue Scoring Rules

Manual reviewer has a score sheet:

| Rule | Score Impact | How It Applies |
|---|---|---|
| Single image only | -1 | Apply when the required second image is missing, because the submission cannot be checked from two views. |
| Missing image description | -1 | Apply when the required second image is missing, because the submission cannot be checked from two views. |
| Duplicate submission | Automatic −1 | A duplicate receives an automatic score of −1. Exception: an accidental duplicate from the same hotkey receives 0 for one copy, while the other copy is reviewed normally. |
| Synthetic image | Automatic −3 | A synthetic image receives an automatic score of −3 because the task explicitly requires real images. |